

PROJECT TOPIC: NATURAL DISASTERS IN MAHARASHTRA & STUDENT ROLE

Topic Description: Comprehensive Study of Regional Natural Disasters Affecting Maharashtra (Floods, Landslides, Droughts, Earthquakes) and Action Plan for Student Risk Reduction

Maharashtra, with its diverse physiography spanning the narrow coastal Konkan plain, the steep rugged Sahyadri mountain ranges (Western Ghats), and the vast semi-arid Deccan plateau, is exposed to unique, recurrent regional natural disaster vulnerabilities. In recent years, unscientific slope cutting for hill roads, deforestation, destruction of natural river floodplains, and extreme monsoon precipitation anomalies have triggered catastrophic disasters across the state.

This project documents an extensive investigative and analytical study of **Natural Disasters Affecting Maharashtra**, profiling four recurring major disaster categories: **Monsoon Riverine & Urban Floods** (Kolhapur/Sangli/Mumbai), **Hilly Landslides & Debris Avalanches** (Malin/Taliye), **Chronic Agrarian Droughts** (Marathwada/Vidarbha), and **Intra-Plate Seismic Tremors** (Killari/Koyna).

The study evaluates the underlying geological, hydrometeorological, and anthropogenic root causes behind these disasters, documents the severe socio-economic impacts on rural agrarian and urban livelihoods, and formulates a concrete, actionable framework defining the proactive role of Standard XI students in disaster preparedness and risk reduction.

2. Importance of the topic (विषयाचे महत्त्व)

Documenting Maharashtra's regional disaster profile is critical for understanding local vulnerability and community safety:

Key Reasons Why State Disaster Profiling is Essential :-

- **Distinct Physiographic Hazard Zones:** Maharashtra's disaster profile is regionally specific: the Konkan and Western Ghats face flash floods and deadly landslides; the rain-shadow Deccan plateau faces devastating multi-year agricultural droughts; and the South Konkan and Marathwada tracts carry intra-plate earthquake risks.
- **Unpacking Anthropogenic Aggravation:** Modern disasters in the state are largely man-made: the tragic 2014 Malin landslide and 2021 Taliye disaster were directly accelerated by heavy machinery hill flattening and deforestation of steep slopes for commercial cash crops.
- **Encroachment on River Floodplains (Blue & Red Lines):** Chronic inundation in Sangli, Kolhapur, and Mumbai occurs because statutory river floodplains and natural mangrove wetlands have been recklessly concretized for residential real estate.
- **Empowering Youth as Grassroots Disaster Mitigators:** Trains junior college students in practical early warning dissemination, flood survival swimming, first-aid triage, and emergency volunteering during state crises.

Disaster risk reduction in Maharashtra demands that development plans strictly respect regional ecological and geological limits.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

The state-level disaster investigation was structured around focused geographical, analytical, and community objectives:

Structured Project Objectives :-

- **Disaster Typology & Regional Mapping:** To compile a comprehensive vulnerability profile of four major natural hazards affecting Maharashtra: Floods, Landslides, Droughts, and Earthquakes.
- **Case Study Documentation:** To examine documented historic state disasters, including the 1993 Killari Earthquake, the 2014 Malin Landslide, the 2019/2021 Krishna River Floods, and Marathwada droughts.
- **Socio-Economic Impact Evaluation:** To quantify losses inflicted on human life, livestock, agricultural crops, rural infrastructure, and state economic productivity.
- **Institutional Response Audit:** To review the disaster response coordination between the Maharashtra State Disaster Management Authority (MSDMA), SDRF, and civil administration.
- **Student Action Plan Formulation:** To define 5 concrete, actionable disaster risk reduction responsibilities that students can execute before, during, and after disasters.

The scope encompasses state disaster management plans, meteorological archives, and published environmental monographs.

4. Project Work Methodology (प्रकल्प कार्यपद्धती)

A standardized secondary documentary research, regional mapping, and analytical synthesis methodology was deployed:

Methodological Research Protocol :-

- **Phase 1 – State Disaster Management Plan Review:** Studied official hazard, vulnerability, and capacity assessment reports published by the Maharashtra State Disaster Management Authority (MSDMA), Mantralaya, Mumbai.
- **Phase 2 – Regional Hazard Zonation Mapping:** Classified Maharashtra's 36 districts into specific hazard vulnerability zones based on seismic maps, flood history, and drought frequency.
- **Phase 3 – Historical Disaster Case Auditing:** Analyzed official inquiry reports and academic post-mortem studies of the Malin (Pune, 2014) and Taliye (Raigad, 2021) landslides and the Kolhapur floods.
- **Phase 4 – Role Definition for Youth:** Synthesized community-based disaster risk management (CBDRM) modules to extract practical student-level responsibilities.
- **Phase 5 – Report Synthesis:** Structured findings according to Chapter 7 of the Maharashtra State Board Standard XI EVS curriculum.

All recorded statistics were verified against government gazettes and state relief and rehabilitation reports.

5. Observations (निरीक्षणे)

Systematic survey of natural disasters affecting Maharashtra reveals distinct regional patterns, impacts, and vulnerability zones:

5.1 Matrix of Major Natural Disasters Affecting Maharashtra

Disaster Category	High-Vulnerability Districts	Underlying Root Causes	Documented Historic Event & Impact
Riverine & Urban Floods	Kolhapur, Sangli, Raigad, Ratnagiri, Mumbai, Thane	Extreme monsoon cloudbursts; backwater effect of dams; river floodplain concrete encroachment.	2019 & 2021 Floods: Panchganga and Krishna rivers submerged Kolhapur/Sangli cities for 10 days; 4.5 lakh people evacuated; massive crop loss.
Hill Slope Landslides	Raigad, Ratnagiri, Pune, Satara, Sindhudurg	Steep Western Ghats slopes; hill cutting for roads; deforestation; heavy prolonged rainfall saturating soil.	Malin (2014) & Taliye (2021): Entire village buried under massive mudslide overnight; 151 deaths in Malin, 86 in Taliye; complete village burial.
Agrarian Droughts	Beed, Osmanabad, Latur, Jalna, Solapur, Ahmednagar	Rain-shadow geographical belt; erratic monsoons; over-extraction of groundwater for sugarcane cash crops.	2012-2016 Drought: Latur required special 'Jaldoot' water trains; extensive crop failure; acute livestock distress and fodder scarcity.
Earthquakes (Seismic)	Latur, Osmanabad, Koyna (Satara), Ratnagiri	Intra-plate fault movements; reservoir-induced seismicity (RIS) 5 Koyna dam.	1993 Killari Earthquake (M 6.2): Shallow focal depth; poorly built stone-mud masonry houses collapsed; over 9,900 deaths.

6. Analysis of data (माहितीचे विश्लेषण)

Analytical examination of Maharashtra's disaster profile illustrates the critical interplay between human action and natural hazards:

Core Analytical Inferences :-

- **The Myth of 'Purely Natural' Disasters:** Analysis reveals that almost every modern disaster in Maharashtra is aggravated by poor land use. The catastrophic Malin landslide occurred because heavy mechanical JCB machinery flattened steep upper hill slopes for agricultural schemes, destroying deep native tree root anchors and turning soil into unstable liquid slurry.
- **River Choking and Dam Mismanagement:** The repeated flooding of Sangli and Kolhapur was triggered not merely by rainfall, but by poor dam discharge coordination between Maharashtra's Koyna/Radhanagari dams and Karnataka's Almatti dam, creating severe backwater stagnation across urban floodplains.
- **The Cash-Crop Thirst Driving Drought:** Marathwada's recurring water crisis is not a rainfall shortage alone, but a crop selection crisis. Growing water-guzzling sugarcane (which consumes 2,500 liters of water per kg of sugar) in a semi-arid rain-shadow belt sucks underground aquifers dry, starving rural families of drinking water.
- **Structural Vulnerability as the Primary Earthquake Killer:** Earthquakes do not kill people; poorly engineered, heavy stone-and-mud buildings with unreinforced masonry roofs kill people, as demonstrated in Killari. Earthquake-resistant construction is an urgent state necessity.

The analysis proves that sustainable land and water management is the most effective disaster prevention strategy.

7. Results and Conclusions (निष्कर्ष)

Conclusion: Maharashtra's recurrent natural disasters demand an urgent transition from reactive post-disaster relief to proactive, long-term disaster risk reduction (DRR). By stopping illegal hill slope cutting, respecting river floodplains, and training student volunteers, the state can protect millions of vulnerable lives.

Concrete Action Plan: The Role of Students in Reducing Disaster Risk :-

- ❑ **Phase 1 (Preparedness):** Prepare a 72-hour family emergency kit (Go-Bag) with essential food, water, and medicines before the onset of the monsoon.
- ❑ **Phase 2 (Early Warning):** Track official MSDMA/IMD alerts and dispel false viral rumors on social media during severe weather emergencies.
- ❑ **Phase 3 (Hazard Reporting):** Report developing cracks on steep hill slopes or unauthorized construction in river blue/red lines to local authorities.
- ❑ **Phase 4 (First-Aid Training):** Complete certified First-Aid and CPR training through Red Cross or Civil Defence corps to act as trained community first responders.
- ❑ **Phase 5 (Post-Disaster Relief):** Volunteer in organized campus relief drives to collect clean drinking water, dry rations, and chlorine purification tablets for affected flood victims.

8. Bibliography / References (संदर्भ सूची)

1. Maharashtra State Disaster Management Authority (MSDMA). (2020). *State Disaster Management Plan*. Mantralaya, Mumbai.
2. Geological Survey of India (GSI). (2015). *Report on the Malin Landslide Disaster, Pune District*. Kolkata.
3. Maharashtra State Board. (2020). *Environmental Education and Water Security (Std. XI)*. Balbharati, Pune.
4. Parasuraman, S., and Unnikrishnan, P. V. (2000). *India Disasters Report: Towards*